```
# This Python 3 environment comes with many helpful analytics
libraries installed
# It is defined by the kaggle/python Docker image:
https://github.com/kaggle/docker-python
# For example, here's several helpful packages to load

import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)

# Input data files are available in the read-only "../input/"
directory
# For example, running this (by clicking run or pressing Shift+Enter)
will list all files under the input directory

import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))

# You can write up to 20GB to the current directory (/kaggle/working/)
that gets preserved as output when you create a version using "Save &
Run All"
# You can also write temporary files to /kaggle/temp/, but they won't
be saved outside of the current session

import tensorflow as tf
print(tf.config.list_physical_devices('GPU'))
```

```
[PhysicalDevice(name='/physical_device:GPU:0', device_type='GPU'),
PhysicalDevice(name='/physical_device:GPU:1', device_type='GPU')]
```

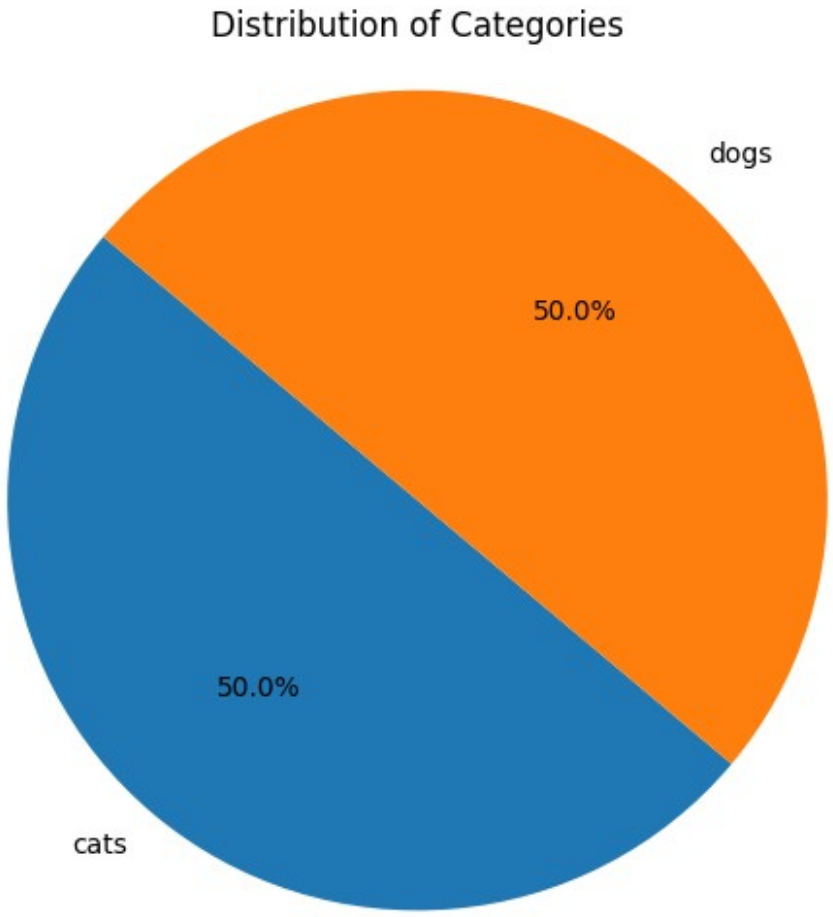
```
import cv2
import os
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import plotly.graph_objects as go
#Matplot Images
import matplotlib.image as mpimg
# Tensflor and Keras Layer and Model and Optimize and Loss
import tensorflow as tf
from tensorflow import keras
from keras import Sequential
from keras.layers import *
from tensorflow.keras.losses import BinaryCrossentropy
# import tensorflow_hub as hub
from tensorflow.keras.optimizers import Adam
#PreTrained Model VGG16
from tensorflow.keras.applications import ResNet50
```

```python
from tensorflow.keras.applications import Xception
#Image Generator DataAugmentation
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.preprocessing import image
#Early Stopping
from tensorflow.keras.callbacks import EarlyStopping
# Warnings Remove
import warnings
warnings.filterwarnings("ignore")

# Directory containing the "Train" folder
directory = "/kaggle/input/dogs-vs-cats"

# List of categories (subfolder names)
categories = ["cats", "dogs"]

# Initialize lists to store filenames and categories
filenames = []
category_labels = []

# Iterate through the categories
for category in categories:
    # Path to the current category folder
    category_folder = os.path.join(directory, "train", category)
    # List all filenames in the category folder
    category_filenames = os.listdir(category_folder)
    # Append filenames and corresponding category labels
    filenames.extend(category_filenames)
    category_labels.extend([category] * len(category_filenames))

# Create DataFrame
df = pd.DataFrame({
    'filename': filenames,
    'category': category_labels
})

# Display the first few rows of the DataFrame
print(df.head())
```

```
        filename category
0  cat.12461.jpg     cats
1  cat.10176.jpg     cats
2   cat.8194.jpg     cats
3   cat.3498.jpg     cats
4    cat.891.jpg     cats
```

```python
# Count the occurrences of each category in the 'category' column
count = df['category'].value_counts()

# Create a pie chart using Seaborn
plt.figure(figsize=(6, 6) , facecolor='white')
```

```
palette = sns.color_palette("tab10")
sns.set_palette(palette)
plt.pie(count, labels=count.index, autopct='%1.1f%%', startangle=140)
plt.title('Distribution of Categories')
plt.axis('equal')

plt.show()  # Show the plot
```




```
def visualize_images(path, num_images=5):
    # Get a list of image filenames in the specified path
    image_filenames = os.listdir(path)

    # Limit the number of images to visualize if there are more than
num_images
    num_images = min(num_images, len(image_filenames))

    # Create a figure and axis object to display images
    fig, axes = plt.subplots(1, num_images, figsize=(15,
3),facecolor='white')
```

```
    # Iterate over the selected images and display them
    for i, image_filename in enumerate(image_filenames[:num_images]):
        # Load the image using Matplotlib
        image_path = os.path.join(path, image_filename)
        image = mpimg.imread(image_path)

        # Display the image
        axes[i].imshow(image)
        axes[i].axis('off')  # Turn off axis
        axes[i].set_title(image_filename)  # Set image filename as
title

    # Adjust layout and display the figure
    plt.tight_layout()
    plt.show()

# Specify the path containing the images to visualize
path_to_visualize = "/kaggle/input/dogs-vs-cats/train/cats"

# Visualize some images from the specified path
visualize_images(path_to_visualize, num_images=5)
```


```
path_to_visualize = "/kaggle/input/dogs-vs-cats/train/dogs"

# Visualize some images from the specified path
visualize_images(path_to_visualize, num_images=5)
```


```
data_dir = '/kaggle/input/dogs-vs-cats/train'

# Defining data generator with Data Augmentation
data_gen_augmented = ImageDataGenerator(rescale = 1/255.,
```

```
                                        validation_split = 0.2,
                                        zoom_range = 0.2,
                                        horizontal_flip= True,
                                        rotation_range = 20,
                                        width_shift_range=0.2,
                                        height_shift_range=0.2)
print('Augmented training Images:')
train_ds = data_gen_augmented.flow_from_directory(data_dir,

target_size = (224, 224),

batch_size = 32,
                                                          subset =
'training',

class_mode = 'binary')

#Testing Augmented Data
# Defining Validation_generator withour Data Augmentation
data_gen = ImageDataGenerator(rescale = 1/255., validation_split =
0.2)

print('Unchanged Validation Images:')
validation_ds = data_gen.flow_from_directory(data_dir,
                                        target_size = (224, 224),
                                        batch_size = 32,
                                        subset = 'validation',
                                        class_mode = 'binary')
```

```
Augmented training Images:
Found 16000 images belonging to 2 classes.
Unchanged Validation Images:
Found 4000 images belonging to 2 classes.
```

```
# Load the pre-trained Xception model without the top (classification)
layer
xception_base = Xception(weights='imagenet', include_top=False,
input_shape=(224, 224, 3))
xception_base.trainable = False
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-
applications/xception/
xception_weights_tf_dim_ordering_tf_kernels_notop.h5
83683744/83683744 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
```

```
# Build the model
model = Sequential()

# Add the pre-trained Xception base
model.add(xception_base)
```

```python
# Add global average pooling layer to reduce spatial dimensions
model.add(AveragePooling2D(pool_size=(2,2)))

# Flatten the feature maps
model.add(Flatten())

# Add a dense layer with 220 units and ReLU activation function
model.add(Dense(220, activation='relu'))

# Add the output layer with 1 unit and sigmoid activation function for binary classification
model.add(Dense(1, activation='sigmoid'))

model.summary()
```

```
Model: "sequential_1"
```

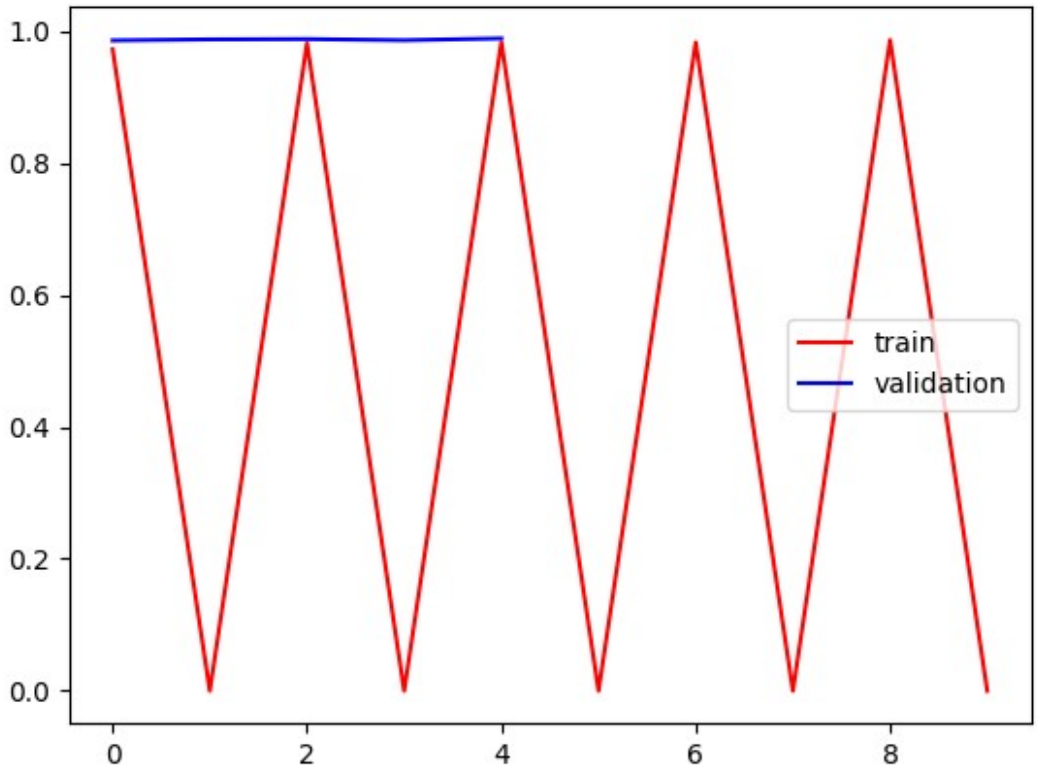
Layer (type)	Output Shape	Param #
xception (Functional)	(None, 7, 7, 2048)	20,861,480
average_pooling2d (AveragePooling2D)	(None, 3, 3, 2048)	0
flatten (Flatten)	(None, 18432)	0
dense (Dense)	(None, 220)	4,055,260
dense_1 (Dense)	(None, 1)	221

```
 Total params: 24,916,961 (95.05 MB)

 Trainable params: 4,055,481 (15.47 MB)

 Non-trainable params: 20,861,480 (79.58 MB)
```

```
#Compile
model.compile(loss = BinaryCrossentropy(),
                optimizer =
keras.optimizers.RMSprop(learning_rate=0.0001, rho=0.9),
                metrics = ['accuracy'])

#Early_Stopping
early_stopping = EarlyStopping(
    min_delta=0.001, # minimium amount of change to count as an
improvement
    patience=5,
    restore_best_weights=True,
)

#Fitting Model
history = model.fit(train_ds,
                        epochs= 10,
                        steps_per_epoch = len(train_ds),
                        validation_data = validation_ds,
                        validation_steps = len(validation_ds),
                        callbacks = early_stopping)
```

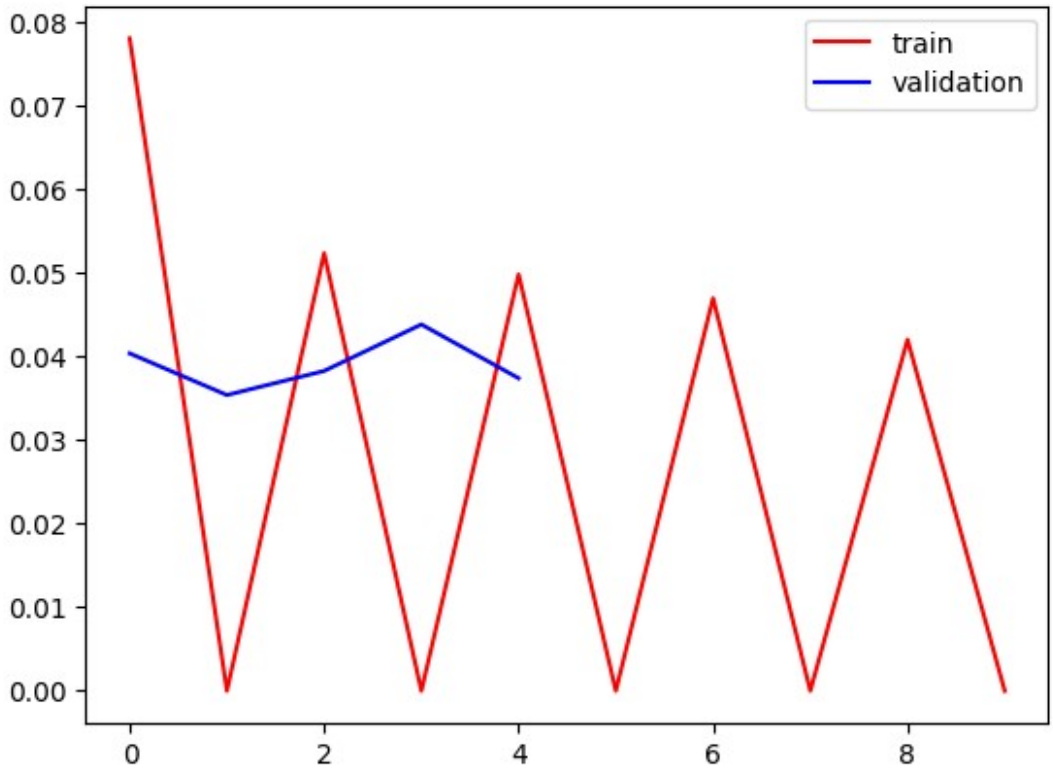
```
Epoch 1/10
500/500 ━━━━━━━━━━━━━━━━━━━━ 361s 680ms/step - accuracy: 0.9582 -
loss: 0.1149 - val_accuracy: 0.9855 - val_loss: 0.0403
Epoch 2/10
500/500 ━━━━━━━━━━━━━━━━━━━━ 0s 81us/step - accuracy: 0.0000e+00 -
loss: 0.0000e+00
Epoch 3/10
500/500 ━━━━━━━━━━━━━━━━━━━━ 206s 408ms/step - accuracy: 0.9816 -
loss: 0.0519 - val_accuracy: 0.9870 - val_loss: 0.0354
Epoch 4/10
500/500 ━━━━━━━━━━━━━━━━━━━━ 0s 27us/step - accuracy: 0.0000e+00 -
loss: 0.0000e+00
Epoch 5/10
500/500 ━━━━━━━━━━━━━━━━━━━━ 208s 411ms/step - accuracy: 0.9825 -
loss: 0.0510 - val_accuracy: 0.9875 - val_loss: 0.0382
Epoch 6/10
500/500 ━━━━━━━━━━━━━━━━━━━━ 0s 38us/step - accuracy: 0.0000e+00 -
loss: 0.0000e+00
Epoch 7/10
500/500 ━━━━━━━━━━━━━━━━━━━━ 210s 415ms/step - accuracy: 0.9809 -
loss: 0.0487 - val_accuracy: 0.9858 - val_loss: 0.0438
Epoch 8/10
500/500 ━━━━━━━━━━━━━━━━━━━━ 0s 35us/step - accuracy: 0.0000e+00 -
loss: 0.0000e+00
Epoch 9/10
500/500 ━━━━━━━━━━━━━━━━━━━━ 205s 403ms/step - accuracy: 0.9864 -
loss: 0.0387 - val_accuracy: 0.9885 - val_loss: 0.0374
Epoch 10/10
```

```
500/500 ━━━━━━━━━━━━━━━━━━━━ 0s 34us/step - accuracy: 0.0000e+00 -
loss: 0.0000e+00

# Evaluate the model on the validation dataset
validation_loss, validation_accuracy = model.evaluate(validation_ds)

# Print the validation loss and accuracy
print("Validation Loss:", validation_loss)
print("Validation Accuracy:", validation_accuracy)

125/125 ━━━━━━━━━━━━━━━━━━━━ 20s 157ms/step - accuracy: 0.9850 - loss:
0.0409
Validation Loss: 0.03535597398877144
Validation Accuracy: 0.9869999885559082

# Accuracy and Val_Accuracy
plt.plot(history.history['accuracy'],color='red',label='train')
plt.plot(history.history['val_accuracy'],color='blue',label='validatio
n')
plt.legend()
plt.show()
```




```
# Loss and Val_Loss
plt.plot(history.history['loss'],color='red',label='train')
plt.plot(history.history['val_loss'],color='blue',label='validation')
```

```
plt.legend()
plt.show()
```




```
# Get the class indices assigned by the generators
class_indices_train = train_ds.class_indices

# Print the class indices
print("Class indices for training generator:", class_indices_train)
```

```
Class indices for training generator: {'cats': 0, 'dogs': 1}
```

```
#Testing Augmented Data
test_dir_path = "/kaggle/input/dogs-vs-cats/test"
# Defining Validation_generator withour Data Augmentation
data_test_gen = ImageDataGenerator(rescale = 1/255.)

print('Test Validation Images:')
test_ds = data_gen.flow_from_directory(test_dir_path,
                                       target_size = (224, 224),
                                       batch_size = 32,
                                       subset = 'validation',
                                       class_mode = 'binary')
```

```
Test Validation Images:
Found 1000 images belonging to 2 classes.
```

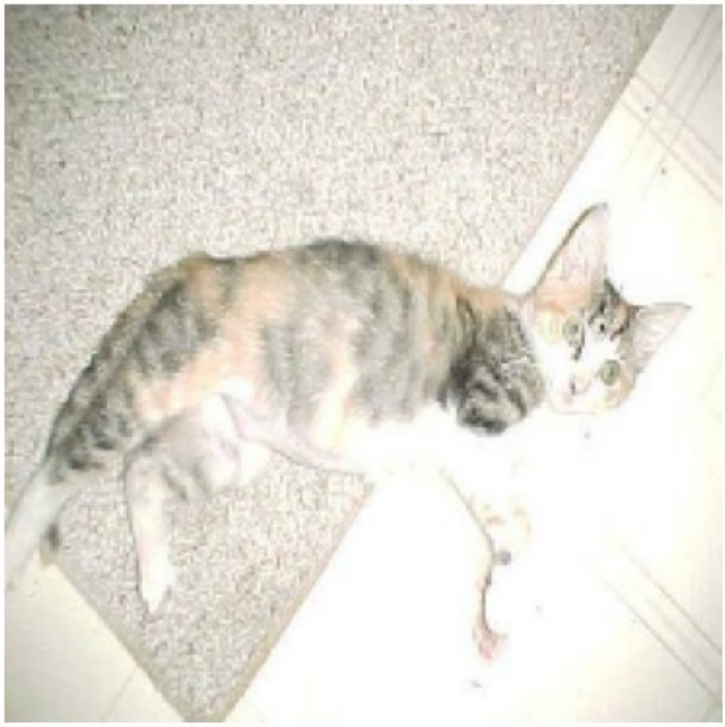
```
# Evaluate the model on the validation dataset
test_loss, test_accuracy = model.evaluate(test_ds)

# Print the validation loss and accuracy
print("Test Loss:", test_loss)
print("Test Accuracy:", test_accuracy)
```

```
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 165ms/step - accuracy: 0.9924 - loss:
0.0273
Test Loss: 0.03659672662615776
Test Accuracy: 0.9900000095367432
```

```
# List of paths to your single images
image_paths = ['/kaggle/input/dogs-vs-cats/train/cats/cat.32.jpg',
'/kaggle/input/dogs-vs-cats/train/cats/cat.44.jpg',
'/kaggle/input/dogs-vs-cats/train/dogs/dog.51.jpg']
# Intialize true labels
true_labels = ['Cat', 'Cat','Dog']

# Load and preprocess each image, make predictions, and display them
using a loop
for img_path, true_label in zip(image_paths, true_labels):
    # Load the image using OpenCV
    img = cv2.imread(img_path)
    # Resize the image to (224, 224)
    img = cv2.resize(img, (224, 224))

    # Normalize pixel values
    img_array = img.astype(np.float32) / 255.0

    # Expand the dimensions to match the input shape expected by the
model
    img_array = np.expand_dims(img_array, axis=0)

    # Make predictions
    predictions = model.predict(img_array)
    actual_prediction = (predictions > 0.5).astype(int)

    # Display the image with true and predicted labels
    # Convert BGR to RGB for displaying with matplotlib
    plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
    plt.axis('off')
    if actual_prediction[0][0] == 0:
        predicted_label = 'Cat'
    else:
        predicted_label = 'Dog'
    plt.title(f'Predicted: {predicted_label} (True: {true_label})')
    plt.show()
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step
```

Predicted: Cat (True: Cat)



```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step
```

Predicted: Cat (True: Cat)

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step
```